Light

Data Representation — Media (Text, Images, Sound)

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Typical pitfalls

OCR GCSE Computer Science — **Data Representation: Characters, Images and Sound (J277/01 Section 1.2.5—1.2.7)**. This topic covers how computers represent text using character sets (ASCII, Unicode), images using pixels (resolution, colour depth), sound using samples (sample rate, bit depth), and how compression reduces file sizes. Exam questions typically ask you to **calculate** file sizes for images and sound, **explain** the effect of changing resolution or sample rate, **compare** ASCII and Unicode, and **describe** the difference between lossy and lossless compression. The 2024 exam included questions about colour depth, character representation, and file size calculations.

What you must know

This checklist covers the key exam-visible concepts. You should be able to explain each point and perform the calculations.

- **Explain** what a character set is and why computers need them.
- **Compare** ASCII (7-bit, 128 characters) and Unicode (up to 32-bit, over 100,000 characters).
- **Describe** how images are represented using pixels, resolution, and colour depth.
- **Calculate** image file size: width × height × colour depth.
- **Explain** how changing resolution or colour depth affects image quality and file size.

- **Describe** how sound is sampled and stored digitally using sample rate and bit depth.
- **Calculate** sound file size: sample rate × bit depth × duration (× channels for stereo).
- **Explain** the difference between lossy and lossless compression.
- **Give examples** of when each compression type is appropriate.

> **Specification note.** OCR J277 Section 1.2.5 covers character representation, 1.2.6 covers images (including metadata), and 1.2.7 covers sound and compression. File size calculations are common exam questions — always show your working and give appropriate units.

Toolkit

Image file size formula

File size (bits) = width (pixels) × height (pixels) × colour depth (bits)
To convert to bytes: divide by 8
To convert to kilobytes: divide by 8, then by 1,000
Example: A 1920 × 1080 image with 24-bit colour depth:
1920 × 1080 × 24 = 49,766,400 bits = 6,220,800 bytes = 6,220.8 KB ≈ 6.2 MB

Sound file size formula

File size (bits) = sample rate (Hz) × bit depth × duration (seconds)
For stereo: multiply by 2 (two channels)
To convert to bytes: divide by 8
Example: 30 seconds of mono audio at 44,100 Hz with 16-bit depth:
44,100 × 16 × 30 = 21,168,000 bits = 2,646,000 bytes = 2,646 KB ≈ 2.6 MB

Character sets comparison

Feature	ASCII	Extended ASCII	Unicode
Bits per character	7 bits	8 bits	8—32 bits (variable)
Number of characters	128	256	Over 140,000
Characters included	English letters, digits, basic punctuation, control characters	ASCII + accented letters, symbols	All world languages, emojis, mathematical symbols, historic scripts
File size impact	Smallest	Small	Larger (but variable)

Colour depth and number of colours

Colour depth	Number of colours	Common name
1 bit	$2^1 = 2$	Black and white
4 bits	$2^4 = 16$	Basic colour
8 bits	$2^8 = 256$	256 colours / GIF
16 bits	$2^{16} = 65{,}536$	High colour
24 bits	$2^{24} = 16{,}777{,}216$	True colour (8 bits per RGB channel)
32 bits	$2^{32} = 4+$ billion	True colour + alpha (transparency)

Compression comparison

Feature	Lossy compression	Lossless compression
Data loss	Yes — some data permanently removed	No — original data fully recoverable
File size reduction	Large (often 90%+ reduction)	Smaller (typically 50-60% reduction)
Quality	Some quality loss (may be unnoticeable)	No quality loss
Example formats	JPEG, MP3, MP4	PNG, FLAC, ZIP
Best for	Photos, music, video streaming	Text, code, medical images, archival

Exam tip: Always show the formula, substitute values, and give units in file size calculations. Even if arithmetic is wrong, you can earn method marks. Convert to sensible units (KB for small files, MB for larger ones).

1. Character representation

Computers store text as numbers. A **character set** is a list that assigns a unique number (code) to each character. When you press a key, the computer stores the corresponding number; when displaying text, it converts numbers back to characters.

1.1 What is a character set?

A **character set** (or character encoding) is a defined list of characters and the binary codes used to represent them. Different character sets exist because different languages and applications have different needs.

Why character sets are needed: Computers only understand binary. To store and display text, we need an agreed system where each character has a specific binary code. Without a standard character set, text would appear as random symbols when transferred between systems.

Characters included: Letters (uppercase and lowercase), digits (0-9), punctuation marks, special symbols (@, #, £), control characters (newline, tab), and whitespace (space).

1.2 ASCII (American Standard Code for Information Interchange)

ASCII is the original standard character set, developed in the 1960s for early computers and telecommunications. It was designed for English-language computing and remains the foundation of modern character encoding.

Key facts about ASCII:

- Uses **7 bits** per character
- Can represent 2^7 = **128 characters**
- Includes uppercase (A-Z), lowercase (a-z), digits (0-9), punctuation, and 33 control characters
- Character codes 0-31 are control characters (non-printable) — these control devices like printers
- Character codes 32-126 are printable characters
- Character code 32 is space, 65 is 'A', 97 is 'a', 48 is '0'

Common ASCII values to know:

Character	Denary code	Notes
Space	32	First printable character
0-9	48-57	Digits in order
A-Z	65-90	Uppercase letters
a-z	97-122	Lowercase letters (32 higher than uppercase)

Extended ASCII uses 8 bits (1 byte) per character, allowing 256 characters. The extra 128 codes (128-255) include accented letters (é, ñ, ü), currency symbols (€, £, ¥), and graphical characters. Different extended ASCII versions exist for different regions (e.g., Western European, Greek, Cyrillic).

> **Exam tip:** The examiner's report notes that many candidates confuse 7-bit ASCII (128 characters) with 8-bit extended ASCII (256 characters). Be precise about which one you're describing.

1.3 Unicode

Unicode is a modern character set designed to represent every character from every language in the world, plus symbols, emojis, and historic scripts. It was developed to

replace the many incompatible regional character sets that existed before.

Key facts about Unicode:

- Uses **variable length encoding** (8, 16, or 32 bits depending on the character)
- The most common encoding is **UTF-8** (1-4 bytes per character)
- Contains over **140,000 characters** (and growing with each version)
- Includes all ASCII characters (codes 0-127 are identical to ASCII)
- Supports Chinese, Japanese, Korean, Arabic, Hebrew, Hindi, Thai, and many more scripts
- Includes emojis, mathematical symbols, musical notation, and historic scripts like Egyptian hieroglyphics

Unicode encoding formats:

- **UTF-8:** Variable length (1-4 bytes). Most common on the web. ASCII-compatible — English text uses 1 byte per character.
- **UTF-16:** Variable length (2 or 4 bytes). Used internally by Windows and Java.
- **UTF-32:** Fixed length (4 bytes per character). Simple but uses more space.

Why Unicode replaced ASCII: ASCII only supports English characters. As computing became global, a standard was needed that could represent all languages. Unicode solves this by providing unique codes for characters from every writing system. A Chinese website and an English website can both use Unicode without garbled text appearing.

Backward compatibility: Unicode was designed to be compatible with ASCII. The first 128 Unicode characters have the same codes as ASCII, so ASCII text is automatically valid UTF-8. This made adoption easier.

1.4 Calculating text file size

The size of a text file depends on the number of characters and the bits per character:

File size (bits) = number of characters × bits per character

Example: A document with 5,000 ASCII characters:

- Using 7-bit ASCII: 5,000 × 7 = 35,000 bits = 4,375 bytes ≈ 4.4 KB
- Using 8-bit extended ASCII: 5,000 × 8 = 40,000 bits = 5,000 bytes = 5 KB

Note: Real text files include additional overhead (headers, formatting), so actual file sizes may be slightly larger.

> **2024 exam context:** Questions asked about the minimum number of bits needed to represent a character set with a given number of characters. Remember: if you need n characters, you need enough bits so that $2^{bits} \geq n$.

Worked Example 1 – Explaining what a character set is

Question: Explain what is meant by a character set. [2 marks]

Model answer:
A character set is a defined list/collection of characters **(1)** where each character is assigned a unique binary code/number **(1)**.

> **Exam tip:** Two key points: (1) it's a list of characters, (2) each has a unique code. Don't just say "it's how text is stored" — be specific about the code-to-

character mapping.

Worked Example 2 — Comparing ASCII and Unicode

Question: Describe two differences between ASCII and Unicode. [4 marks]

Model answer:
1. ASCII uses 7 bits per character **(1)**, whereas Unicode uses variable length encoding (8-32 bits per character) **(1)**.
2. ASCII can represent 128 characters (English letters, digits, symbols) **(1)**, whereas Unicode can represent over 140,000 characters including all world languages and emojis **(1)**.

Exam tip: For "describe differences" questions, make sure each point covers BOTH ASCII AND Unicode. A single statement like "ASCII uses 7 bits" only scores half marks.

Worked Example 3 — Minimum bits for a character set

Question: A character set contains 150 characters. What is the minimum number of bits needed to represent each character? [2 marks]

Method:
Find the smallest power of 2 that is ≥ 150:

- 2^7 = 128 (not enough — 128 < 150)
- 2^8 = 256 (enough — 256 ≥ 150)

Answer: **8 bits**

Exam tip: The question asks for MINIMUM bits. Show that the previous power of 2 isn't enough, then show that your answer is sufficient.

Worked Example 4 — Why Unicode is needed

Question: Explain why Unicode was developed to replace ASCII. [2 marks]

Model answer:
ASCII can only represent English characters and basic symbols **(1)**. Unicode was developed to represent characters from all world languages (Chinese, Arabic, Hindi, etc.) and additional symbols like emojis **(1)**.

> **Exam tip:** Focus on the limitation of ASCII (English only) and how Unicode solves it (all languages). Don't just say "Unicode has more characters" — explain WHY more characters are needed.

2. Image representation

Digital images are made up of tiny dots called **pixels**. Each pixel has a colour value stored as binary. The quality and file size of an image depend on its **resolution** (number of pixels) and **colour depth** (bits per pixel).

2.1 Pixels and resolution

A **pixel** (picture element) is the smallest unit of a digital image — a single coloured dot. When you zoom in on an image, you eventually see individual pixels as small squares.

Resolution is the number of pixels in an image, usually expressed as width × height (e.g., 1920 × 1080). Higher resolution means more pixels, which gives:

- More detail and sharper images
- Larger file sizes
- Ability to print at larger sizes without blur

Common resolutions:

- HD: 1280 × 720 (921,600 pixels)
- Full HD: 1920 × 1080 (2,073,600 pixels)
- 4K: 3840 × 2160 (8,294,400 pixels)

2.2 Colour depth

Colour depth (or bit depth) is the number of bits used to store the colour of each pixel. More bits = more possible colours.

With n bits of colour depth, you can represent 2^n different colours:

- 1-bit: 2 colours (black and white)
- 8-bit: 256 colours
- 24-bit: 16.7 million colours ("true colour")

True colour (24-bit) uses 8 bits each for Red, Green, and Blue (RGB). Each colour channel can have 256 intensity levels (0-255), giving 256 × 256 × 256 = 16,777,216 possible colours.

> **Exam tip:** The 2024 exam asked about colour depth and the number of colours. Remember: number of colours = $2^{\text{colour depth}}$. If colour depth is 7 bits, maximum colours = 2^7 = 128.

2.3 Calculating image file size

Formula: File size = width × height × colour depth

Step-by-step method:

1. Calculate total pixels: width × height
2. Multiply by colour depth (bits per pixel)
3. Result is in bits — divide by 8 for bytes
4. Convert to appropriate units (KB, MB)

Example: Calculate the file size of a 800 × 600 image with 24-bit colour depth.

- Pixels: 800 × 600 = 480,000

- Bits: 480,000 × 24 = 11,520,000 bits
- Bytes: 11,520,000 ÷ 8 = 1,440,000 bytes
- KB: 1,440,000 ÷ 1,000 = 1,440 KB = 1.44 MB

2.4 Effect of changing resolution and colour depth

Increasing resolution:

- More pixels → more detail
- Larger file size
- Requires more storage and bandwidth

Increasing colour depth:

- More colours → smoother gradients, more realistic
- Larger file size
- More accurate colour representation

Trade-off: Higher quality images take more storage space and longer to transmit. For web use, lower resolution/colour depth may be acceptable to reduce loading times.

2.5 Metadata

Metadata is additional data stored with an image that describes properties of the file but isn't part of the visible image itself. It's often called "data about data".

Examples of image metadata:

- Dimensions (width, height in pixels)
- Colour depth and colour space
- File format and compression type
- Date and time taken
- Camera make and model
- Camera settings (ISO, aperture, shutter speed, focal length)
- GPS location coordinates (if enabled)
- Author/copyright information
- Thumbnail preview

Why metadata matters: Metadata helps organize photos (search by date, location), preserves camera settings for learning, and tracks copyright. However, GPS metadata can be a privacy concern — sharing photos online may reveal your location.

2.6 Common image file formats

Format	Compression	Best for
JPEG	Lossy	Photos, web images where some quality loss is acceptable
PNG	Lossless	Graphics with sharp edges, transparency, screenshots
GIF	Lossless	Simple animations, limited to 256 colours
BMP	None (uncompressed)	Maximum quality, very large files
TIFF	Optional lossless	Professional photography, printing, archival
RAW	None	Professional cameras, maximum editing flexibility

Worked Example 5 — Calculating image file size

Question: Calculate the file size in kilobytes of an image that is 1024 × 768 pixels with a colour depth of 24 bits. Show your working. [3 marks]

Working:
Total pixels = 1024 × 768 = 786,432 pixels **(1)**
File size in bits = 786,432 × 24 = 18,874,368 bits
File size in bytes = 18,874,368 ÷ 8 = 2,359,296 bytes **(1)**
File size in KB = 2,359,296 ÷ 1,000 = **2,359.3 KB** (or 2.36 MB) **(1)**

Exam tip: Show each step clearly. Even if your arithmetic is wrong, you get method marks for correct formula and process. Always include units in your final answer.

Worked Example 6 — Effect of changing colour depth

Question: Explain how reducing the colour depth of an image from 24 bits to 8 bits would affect the image. [4 marks]

Model answer:
The file size would decrease **(1)** because fewer bits are used to store each pixel (8 instead of 24, so file is 1/3 the size) **(1)**.
The image quality would decrease **(1)** because fewer colours are available (256 instead of 16.7 million), causing visible banding in gradients and less accurate colour representation **(1)**.

Exam tip: Always address BOTH file size AND quality when asked about changing resolution or colour depth. Explain the cause (fewer bits/colours) and the effect.

Worked Example 7 — Effect of changing resolution

Question: A photographer increases the resolution of images from 1920 × 1080 to 3840 × 2160. Describe two effects this will have. [4 marks]

Model answer:
1. The file size will increase **(1)** because there are 4 times as many pixels (the resolution has doubled in each dimension), so 4 times as much data needs to be stored **(1)**.
2. The image quality/detail will improve **(1)** because more pixels capture finer detail, allowing larger prints and more detail when zooming **(1)**.

Exam tip: When resolution doubles in BOTH dimensions (1920→3840 and 1080→2160), total pixels QUADRUPLES (2×2=4). Don't say "file size doubles" — it quadruples.

Worked Example 8 — Describing metadata

Question: Describe what is meant by metadata and give two examples of metadata that might be stored with a digital photograph. [3 marks]

Model answer:
Metadata is data about data / additional information stored with a file that describes its properties **(1)**.
Examples: Date and time the photo was taken **(1)**; GPS location where the photo was taken **(1)**.
Other valid examples: Camera model, resolution, colour depth, file format, exposure settings, author/copyright.

Exam tip: "Data about data" is the classic definition of metadata. Examples should be ABOUT the image (date, location, camera) not PART of the image (pixel colours).

3. Sound representation and compression

Sound is an analogue signal — a continuous wave of pressure changes in the air. To store sound digitally, we must **sample** the wave at regular intervals and convert each sample to a binary value. **Compression** reduces file sizes for storage and transmission.

3.1 How sound is digitised

Sampling is the process of measuring the amplitude (height) of a sound wave at regular intervals. Each measurement becomes a number that can be stored in binary.

The analogue-to-digital process:

1. A **microphone** converts sound waves (pressure changes in air) into an electrical signal (analogue)
2. An **ADC (Analogue-to-Digital Converter)** samples the electrical signal at regular intervals
3. Each sample's amplitude is measured and converted to a binary number
4. The sequence of binary values is stored as a digital audio file

The digital-to-analogue process (playback):

1. A **DAC (Digital-to-Analogue Converter)** reads the binary values
2. It reconstructs an electrical signal based on the sample values
3. The electrical signal drives a **speaker**, which converts it back to sound waves

Why sampling works: If we take enough samples close together, the reconstructed wave is almost identical to the original. The more samples we take (higher sample rate)

and the more accurately we measure each sample (higher bit depth), the closer the digital version is to the original analogue sound.

> **Key insight:** Digital audio is an approximation of the original continuous sound wave. The quality of this approximation depends on sample rate and bit depth.

3.2 Sample rate

Sample rate (or sampling frequency) is the number of samples taken per second, measured in Hertz (Hz). One Hz means one sample per second, so 44,100 Hz means 44,100 samples are taken every second.

Effect of sample rate:

- Higher sample rate → more samples per second → captures more detail → more accurate sound reproduction
- Higher sample rate → larger file size (doubling sample rate doubles file size)
- Too low a sample rate causes **aliasing** — distortion where high frequencies are incorrectly recorded as lower frequencies

Common sample rates:

- 8,000 Hz — telephone quality (voice only, bandwidth limited)
- 22,050 Hz — radio quality (acceptable for speech and simple music)
- 44,100 Hz — CD quality (standard for music)
- 48,000 Hz — DVD/professional video quality
- 96,000 Hz — high-resolution audio (audiophile recordings)

Nyquist theorem: To accurately capture a frequency, the sample rate must be at least double that frequency. Humans hear up to approximately 20,000 Hz, so CD quality (44,100 Hz) captures the full audible range with a small safety margin. This is why 44,100 Hz became the standard for music CDs.

3.3 Bit depth

Bit depth (or sample resolution) is the number of bits used to store each sample. It determines how accurately the amplitude can be recorded.

How bit depth affects accuracy: Each sample must be rounded to the nearest available level. With more bits, there are more levels, so rounding errors are smaller. This gives a more accurate representation of the original sound.

Effect of bit depth:

- Higher bit depth → more possible amplitude levels → more accurate sound → better dynamic range
- Higher bit depth → larger file size (doubling bit depth doubles file size)
- Low bit depth can cause **quantisation noise** — a grainy, distorted sound caused by rounding errors

Common bit depths:

- 8-bit: 2^8 = 256 amplitude levels (low quality, grainy sound, small files)
- 16-bit: 2^{16} = 65,536 levels (CD quality, suitable for most purposes)
- 24-bit: 2^{24} = 16.7 million levels (professional/studio quality, more headroom for editing)
- 32-bit: Used for professional audio processing, not typically for final files

Dynamic range: Bit depth affects dynamic range — the difference between the quietest and loudest sounds that can be recorded. 16-bit audio has a dynamic range of about 96 dB (roughly the range of human hearing), while 24-bit offers about 144 dB.

3.4 Calculating sound file size

Formula: File size = sample rate × bit depth × duration (seconds)

For stereo, multiply by 2 (two audio channels).

Example: Calculate the file size for 60 seconds of stereo audio at 44,100 Hz with 16-bit depth.

- Bits = 44,100 × 16 × 60 × 2 = 84,672,000 bits
- Bytes = 84,672,000 ÷ 8 = 10,584,000 bytes
- MB = 10,584,000 ÷ 1,000,000 = **10.58 MB**

> **Exam tip:** Check whether the question says "mono" or "stereo". Stereo has 2 channels, so multiply by 2. If not specified, assume mono (1 channel).

3.5 Lossy compression

Lossy compression reduces file size by permanently removing some data. The removed data cannot be recovered — the compression is irreversible.

How lossy image compression works (JPEG):

- Analyses the image for areas where the human eye won't notice small changes
- Reduces colour precision in areas of similar colour
- Simplifies fine details and textures
- Groups pixels into blocks and averages colours within each block

How lossy audio compression works (MP3):

- Removes frequencies outside normal human hearing range (below 20 Hz, above 20 kHz)
- Uses **psychoacoustic modelling** — removes sounds that are "masked" by louder sounds
- Reduces precision for very quiet sounds
- Groups similar frequencies together

Compression quality settings: Lossy formats typically offer quality settings. Higher quality = larger file, less noticeable loss. Lower quality = smaller file, more visible/audible artefacts.

Advantages:

- Very large file size reductions (often 90%+ for images, 70-90% for audio)
- Faster to transmit and download
- Quality loss may be imperceptible at moderate compression levels
- Ideal for streaming where bandwidth is limited

Disadvantages:

- Quality degrades with each save/re-compression (generation loss)
- Original data cannot be recovered
- Artefacts visible/audible at high compression (blocky images, "underwater" sound)
- Not suitable where accuracy is critical

Common lossy formats: JPEG (images), MP3 and AAC (audio), MP4/H.264/HEVC (video)

Best for: Photos for web/social media, streaming music, video streaming, when storage/bandwidth is limited and some quality loss is acceptable.

3.6 Lossless compression

Lossless compression reduces file size without losing any data. The original file can be perfectly reconstructed from the compressed version.

Common lossless techniques:

- **Run-length encoding (RLE):** Replaces repeated values with a count. "AAAABBCC" becomes "4A2B2C". Works well for images with large areas of the same colour.
- **Dictionary encoding:** Builds a dictionary of common patterns and replaces them with shorter codes. Used in ZIP and PNG.
- **Huffman coding:** Uses shorter codes for more frequent values and longer codes for rare values.

Advantages:

- No quality loss — original data fully recoverable
- Can be compressed and decompressed repeatedly without degradation
- Essential for data where accuracy is critical

Disadvantages:

- Smaller file size reduction than lossy (typically 50-60%)
- Compressed files still relatively large
- Compression ratio varies depending on data patterns

Common lossless formats: PNG (images), FLAC (audio), ZIP (general files), GIF (256-colour images)

Best for: Text files, program code, spreadsheets, medical images, archival of original photos, professional audio editing, legal documents — anywhere data integrity is essential.

3.7 Common audio file formats

Format	Compression	Typical use
WAV	None (uncompressed)	Professional audio, maximum quality
MP3	Lossy	Music files, streaming, portable devices
AAC	Lossy	iTunes, YouTube, better quality than MP3
FLAC	Lossless	Audiophile music, archival
OGG	Lossy	Open-source alternative to MP3

Worked Example 9 — Calculating sound file size

Question: Calculate the file size in megabytes of a 3-minute mono audio recording with a sample rate of 44,100 Hz and a bit depth of 16 bits. Show your working. [4 marks]

Working:
Duration in seconds = 3 × 60 = 180 seconds **(1)**
File size in bits = 44,100 × 16 × 180 = 127,008,000 bits **(1)**
File size in bytes = 127,008,000 ÷ 8 = 15,876,000 bytes **(1)**
File size in MB = 15,876,000 ÷ 1,000,000 = **15.88 MB (1)**

Exam tip: Convert duration to seconds first. Remember the formula: sample

rate × bit depth × duration. For stereo, add "× 2" for the two channels.

Worked Example 10 — Effect of sample rate on sound quality

Question: Explain how increasing the sample rate affects the quality and file size of a digital audio recording. [4 marks]

Model answer:
Increasing sample rate improves sound quality **(1)** because more samples are taken per second, so the digital recording more accurately represents the original analogue sound wave **(1)**.
Increasing sample rate increases file size **(1)** because more samples means more data to store (e.g., doubling sample rate doubles the file size) **(1)**.

Exam tip: Cover BOTH quality AND file size. Explain the relationship (more samples = more accurate representation) rather than just stating the effect.

Worked Example 11 — Comparing lossy and lossless compression

Question: Compare lossy and lossless compression. [4 marks]

Model answer:
Lossy compression permanently removes data **(1)**, whereas lossless compression preserves all original data which can be fully recovered **(1)**.
Lossy compression achieves larger file size reductions (often 90%+) **(1)**, whereas lossless typically achieves smaller reductions (around 50-60%) **(1)**.

Exam tip: For "compare" questions, make sure each point addresses BOTH types. Use comparison words like "whereas", "however", "but" to show the contrast.

Worked Example 12 — Choosing compression type

Question: A hospital stores X-ray images. Explain why lossless compression would be more appropriate than lossy compression for these images. [3 marks]

Model answer:
Lossless compression preserves all the original data without any loss **(1)**.
Medical images like X-rays need to show fine details that could be important for diagnosis **(1)**.

Lossy compression might remove subtle details that could affect medical decisions/diagnosis **(1)**.

Exam tip: Always relate your answer to the CONTEXT (here: medical images, diagnosis). Generic answers about "quality" won't score as well as specific reasons why quality matters HERE.

Worked examples summary

The 12 worked examples above cover the main question types you will encounter on data representation. Here is a summary of the key question patterns:

Example	Question type	Key skill
1	Explain term (2 marks)	Define character set with code mapping
2	Compare ASCII/Unicode (4 marks)	Bits per character + number of characters
3	Minimum bits calculation (2 marks)	Find smallest $2^n \geq$ required characters
4	Explain Unicode need (2 marks)	ASCII limitation → Unicode solution
5	Image file size (3 marks)	width × height × colour depth → KB
6	Effect of colour depth (4 marks)	File size AND quality effects
7	Effect of resolution (4 marks)	File size AND quality effects
8	Describe metadata (3 marks)	Definition + examples
9	Sound file size (4 marks)	sample rate × bit depth × duration
10	Effect of sample rate (4 marks)	Quality AND file size effects
11	Compare compression (4 marks)	Data loss + size reduction comparison
12	Justify compression choice (3 marks)	Apply to specific context

Quick reference

Key terms — exam-ready definitions

Term	Definition (mark scheme style)
Character set	A defined list of characters where each is assigned a unique binary code

Term	Definition (mark scheme style)
ASCII	7-bit character set with 128 characters; includes English letters, digits, basic punctuation
Unicode	Variable-length character set with 140,000+ characters; supports all world languages and symbols
Pixel	The smallest unit of a digital image; a single coloured dot
Resolution	The number of pixels in an image, expressed as width × height
Colour depth	The number of bits used to store the colour of each pixel
Metadata	Data about data; additional information stored with a file describing its properties
Sample rate	The number of audio samples taken per second, measured in Hertz (Hz)
Bit depth (audio)	The number of bits used to store each audio sample
Sampling	Measuring the amplitude of a sound wave at regular intervals to convert it to digital
ADC	Analogue-to-Digital Converter; hardware that converts analogue signals to digital data
Lossy compression	Compression that permanently removes data; original cannot be recovered; large file reduction
Lossless compression	Compression that preserves all data; original fully recoverable; smaller file reduction

File size formulas

Media type	Formula (result in bits)
Text	Number of characters × bits per character
Image	Width × height × colour depth
Sound (mono)	Sample rate × bit depth × duration (seconds)
Sound (stereo)	Sample rate × bit depth × duration × 2

Unit conversions

Conversion	Operation
Bits → Bytes	÷ 8

Conversion	Operation
Bytes → KB	÷ 1,000
KB → MB	÷ 1,000
MB → GB	÷ 1,000

Powers of 2 (colours / character sets)

Bits	1	2	3	4	5	6	7	8	16	24
Values	2	4	8	16	32	64	128	256	65,536	16.7M

Typical pitfalls

Pitfall 1 — Confusing ASCII bit sizes.
Wrong: "ASCII uses 8 bits and has 256 characters."
Right: Standard ASCII uses **7 bits** (128 characters). **Extended** ASCII uses 8 bits (256 characters).

Pitfall 2 — Forgetting to convert minutes to seconds.
Wrong: File size = 44,100 × 16 × 3 (for 3 minutes)
Right: Convert to seconds first: 3 minutes = **180 seconds**, then 44,100 × 16 × 180.

Pitfall 3 — Forgetting stereo has 2 channels.
Wrong: Stereo file size = sample rate × bit depth × duration
Right: Stereo needs **× 2** because there are two audio channels (left and right).

Pitfall 4 — Not converting bits to bytes correctly.
Wrong: Dividing by 10 or 1,000 to convert bits to bytes.
Right: Divide by **8** (8 bits = 1 byte). Then divide by 1,000 for KB.

Pitfall 5 — One-sided comparison answers.
Wrong: "Lossy compression removes data." (only describes lossy)
Right: "Lossy compression removes data, **whereas** lossless preserves all data."

Pitfall 6 — Confusing resolution doubling effect.
Wrong: "Doubling resolution doubles file size."
Right: Doubling resolution in BOTH dimensions **quadruples** file size (2× width × 2× height = 4× pixels).

Pitfall 7 — Vague quality explanations.
Wrong: "Higher colour depth makes better quality."
Right: "Higher colour depth allows **more colours**, so gradients are smoother and colours more accurate."

Pitfall 8 — Not relating compression choice to context.
Wrong: "Lossless is better because it keeps all data."
Right: "Lossless is needed here because **medical diagnosis requires seeing all details**."

Pitfall 9 — Confusing metadata with image data.
Wrong: "Metadata includes the colours of the pixels."
Right: Metadata is data ABOUT the image (date, location, camera), not the image content itself.

Pitfall 10 — Not showing working in calculations.
Wrong: Just writing "2,359 KB" with no steps.
Right: Show formula, substitution, intermediate steps, unit conversions. Method marks are available!

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